

envs-193ds-spring-2026-final

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Setup

```
library(tidyverse)
library(here)
library(janitor)
nests <- read_csv(here("data", "nest_data_final.csv"))
```

Problem 1. Research writing

a. Transparent statistical methods

In part 1, they used a logistic regression to predict the probability that American Avocets would use microhabitats based on distance to water edge.

In part 2, they used a chi-square test to determine if there is a relationship between bird species and habitat type.

b. Suggestions for rewriting

American Avocets were more likely to use some microhabitats more than others based on distance to water edge. The habitat's distance to water edge seems to have an influence on probability that American Avocets would use it.

Habitat use differed between bird species (American Avocet, Forster's Tern, Black-necked Stilt), based on its type (managed pond, non-tidal marsh, salt pond, tidal marsh, tidal mudflat). The bird species were not evenly distributed among these habitat types.

c. Further analysis

Another variable that could influence the probability of American Avocet microhabitat use is distance to nearest road. This would be a continuous variable measured in meters to the nearest road that regular cars drive on. The noise and air pollution from the cars might disrupt and discourage the birds from using the microhabitat, meaning probability of microhabitat use would most likely increase the further the habitat is from a road.

Problem 2. Data analysis

a. Clean and wrangle

```
# creating new object from data
nests_clean <- nests |>
  # selecting columns of interest
  select(height_cm, case_control, ant_species) |>
  # filtering for species of interest
  filter(ant_species == c("RRB", "AB", "BBR")) |>
  # adding new column with ant scientific name
  mutate(scientific_name = recode_values(ant_species,
    "AB" ~ "Crematogaster sjostedti",
    "RRB" ~ "Crematogaster mimosae",
    "BBR" ~ "Crematogaster nigriceps"
  )) |>
  # reordering levels of ant species
  mutate(ant_species = fct_relevel(ant_species,
    "AB",
    "RRB",
    "BBR"))
```

```
# displaying 10 random rows from table
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 4
  height_cm case_control ant_species scientific_name
  <dbl>      <dbl> <fct>      <chr>
1     192          0 RRB      Crematogaster mimosae
2     201          1 BBR      Crematogaster nigriceps
3     157          0 RRB      Crematogaster mimosae
```

4	260.	0 RRB	Crematogaster mimosae
5	114.	0 AB	Crematogaster sjostedti
6	163	0 BBR	Crematogaster nigriceps
7	187	0 RRB	Crematogaster mimosae
8	432.	0 RRB	Crematogaster mimosae
9	156.	0 RRB	Crematogaster mimosae
10	238.	1 RRB	Crematogaster mimosae

b. Response variable

In the `case_control` column, which is the response variable, the 1s represent a tree that has a bird nest in it while the 0s represent trees that did not have a bird nest in it.

c. Purpose of study

Tree height is a continuous variable measured in cm. As tree height increases, the probability of nest occurrence increases because the canopy is denser higher up and provides more cover for the bird nest. This is described in the second paragraph in the Discussion section.

Acacia ant species is a categorical variable, where its presence on a host tree is denoted by a code unique to its species. Based on biological relationships, trees with *C. nigriceps* and *C. mimosae* increase the probability of nest occurrence much more than trees with *C. sjostedti* because they defend their trees from herbivores and bird predators much more aggressively. This is described in the first and second paragraphs in the Discussion section.

d. Table of models

2 models total:

Model number	Tree Height	Ant Species	Model Description
1	X	X	all predictors (no interaction term)
2	X	X	all predictors (with interaction term)

- e. Run the models
- f. Select the best model
- g. Check the diagnostics
- h. Visualize the model predictions
- i. Write a caption for your figure
- j. Calculate model predictions
- k. Interpret your results

Problem 3. Affective and exploratory visualizations